

Kevin Amador Ayala Vilchis

(915) 332-6515 | kevinayalavil74@gmail.com | in/kevin-ayalavilchis | github/KevinVilchisAy |
Personal Website: <https://kevin-ayala-vilchis-personal-websites.netlify.app/>

EDUCATION

Bachelor of Science in Computer Science

The University of Texas at El Paso (UTEP)

Anticipated: May 2028

Overall GPA: 3.72/4.00

Awards: Fall 2023 – Fall 2025 Engineering Dean's List, Engineering Honors Program Alumni, NSF S-STEM 2025 Scholar

Relevant Coursework: Advanced Object-Oriented Programming, Data Structures and Algorithms, Discrete Structures, Calculus II

SKILLS & CERTIFICATIONS

- Programming Languages: Python, Java, SQL, TypeScript
- Data Science & Machine Learning: scikit-learn (Random Forest, Model Evaluation), Pandas, Numpy, Pipelines
- Tools & Technologies: Git, Github, Jupyter Notebooks, Visual Studio, Linux, Tableau
- Web Development: HTML, CSS, JavaScript, Front-End Development

IBM – Tools for Data Science

Sept 2025

- Gained knowledge of core data science concepts (Big Data, Data Mining, Deep Learning) strengthening analytical skills
- Gained hands-on experience with key data science tools, including Jupyter Notebooks, R-Studio & GitHub utilized for developing proficiency in data-driven workflows & collaborations

PROJECTS

Machine Learning Pipeline for Music Popularity Forecasting

Feb 2026 – Feb 2026

- Developed an end-to-end **machine learning pipeline** in Python to predict song popularity (0–100 scale) using 114K+ tracks; trained a **Random Forest Regressor (100 trees)** achieving 55% R^2 & reducing prediction error to 14 RMSE.
- Engineered and standardized 10+ audio features using **Pandas, NumPy, and scikit-learn**, and conducted feature importance analysis identifying danceability, energy, & valence as top drivers of popularity, improving model interpretability by 100%.

TicketMiner System – Full-Stack Object Programming Oriented System

Aug 2025 – Nov 2025

- Developed a **scalable Java-based** ticketing platform with a menu-driven console UI, **supporting 100+ customers and admin workflows**, automated ticket purchases, invoicing, and persistent CSV-based data storage, **reducing manual processing time by ~60% through automation**
- Applied advanced OOP principles & design patterns (**Factory Method, Abstract Factory, MVC**) to design extensible event & venue pipelines, **improving code reusability & system scalability by ~40%**

Solar Energy Predictive Modeling - GMIS Data Analytics Competition

Oct 2025 – Oct 2025

- Programmed a **Python-based** solution as part of a collaborative team of 3 to predict solar panel DC output using weather data sets, **applying linear regression modeling with pandas and scikit-learn, reducing prediction time by 20%**
- Standardized and preprocessed large CSV datasets with **StandardScaler**, enhancing model accuracy and robustness for real-world energy data

Personal Website Portfolio

May 2025 – Jul 2025

- Built a responsive personal portfolio website using **HTML, CSS, & JavaScript**, optimizing assets and layout to improve mobile responsiveness and **reduce page load time by 30%**

WORK EXPERIENCE

Technician - College of Engineering Dean's Office Undergrad Assistant II

Aug 2025 – Current

- Provided technical support for engineering students and faculty (Electrical, Aerospace, etc.), **troubleshooting hardware/software** issues and **reducing solution time by 30%**
- **Managed SQL-based databases** tracking equipment, software licenses & service requests, **increasing accuracy & retrieval efficiency by 40%**

Technician & Webmaster - Department of Computer Science Undergrad Assistant I

Jan 2024 – May 2025

- Solved 50+ monthly technical requests, **reducing resolution time by 30%** & improving overall satisfaction of staff & students
- **Redesigned & optimized** departmental websites using **Cascade CMS, HTML & CSS**, improving site usability & **reducing navigation issues by 40%**

Production Assistant Internship - ABC 7, KVIA News – El Paso Texas Department

Aug 2024 – Dec 2024

- **Curated, edited, & published daily website stories**, including **HTML/CSS** updates & layout changes, while ensuring adherence to editorial standards
- Collaborated with a team of 8 engineers for live broadcasts, **reducing audio issues by 20% of 10+ broadcasts**

LEADERSHIP EXPERIENCE

Mentorship Program Coordinator – Google Developer Group

Jan 2026 – Current

- Coordinate a mentorship program for 20+ mentors and 50+ mentees, helping computer science students build **successful tech careers** and driving **35% higher engagement** through structured check-ins
- **Lead recruitment and mentor-mentee matching**, maintaining accurate records and using feedback to improve **career readiness and program satisfaction by 25%**

Webmaster Chair – Web-App Development Club

Mar 2025 – Current

- Spearheading the design & development of the club's new website using **Cascade CMS, HTML, CSS & JavaScript**, **improving user experience & navigation efficiency by 40%** Collaborating with club officers to restructure site content, **resulting in a 30% increase in user engagement & better alignment with organizational branding & goals**